

3.2 The Derivative as a Function Worksheet

1. Find the derivative of the function using the definition of the derivative. State the domain of the function and the domain of its derivative.

a) $f(x) = \frac{1}{2}x - \frac{1}{3}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\left[\frac{1}{2}(x+h) - \frac{1}{3}\right] - \left[\frac{1}{2}x - \frac{1}{3}\right]}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{2}h}{h} = \lim_{h \rightarrow 0} \frac{1}{2} = \frac{1}{2}$$

$f(x)$ has a domain of $(-\infty, \infty)$.

$f'(x)$ has a domain of $(-\infty, \infty)$.

b) $f(x) = \sqrt{9-x}$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{9-(x+h)} - \sqrt{9-x}}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{9-(x+h)} - \sqrt{9-x}}{h} \cdot \frac{\sqrt{9-(x+h)} + \sqrt{9-x}}{\sqrt{9-(x+h)} + \sqrt{9-x}} \\ &= \lim_{h \rightarrow 0} \frac{9-(x+h) - (9-x)}{h[\sqrt{9-(x+h)} + \sqrt{9-x}]} = \lim_{h \rightarrow 0} \frac{-h}{h[\sqrt{9-(x+h)} + \sqrt{9-x}]} = \lim_{h \rightarrow 0} \frac{-1}{\sqrt{9-(x+h)} + \sqrt{9-x}} = \frac{-1}{2\sqrt{9-x}} \end{aligned}$$

$f(x)$ has a domain of $(-\infty, 9]$.

$f'(x)$ has a domain of $(-\infty, 9)$.

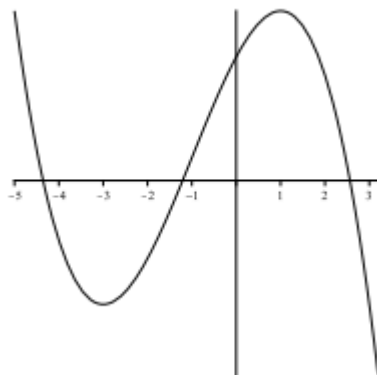
c) $f(x) = x^{3/2}$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^{3/2} - x^{3/2}}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{(x+h)^3} - \sqrt{x^3}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{(x+h)^3} - \sqrt{x^3}}{h} \cdot \frac{\sqrt{(x+h)^3} + \sqrt{x^3}}{\sqrt{(x+h)^3} + \sqrt{x^3}} = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h[\sqrt{(x+h)^3} + \sqrt{x^3}]} \\ &= \lim_{h \rightarrow 0} \frac{[x^3 + 3x^2h + 3xh^2 + h^3] - x^3}{h[\sqrt{(x+h)^3} + \sqrt{x^3}]} = \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h[\sqrt{(x+h)^3} + \sqrt{x^3}]} = \lim_{h \rightarrow 0} \frac{3x^2 + 3xh + h^2}{\sqrt{(x+h)^3} + \sqrt{x^3}} \\ &= \frac{3x^2}{2\sqrt{x^3}} = \frac{3x^2}{2x^{3/2}} = \frac{3}{2}\sqrt{x} \end{aligned}$$

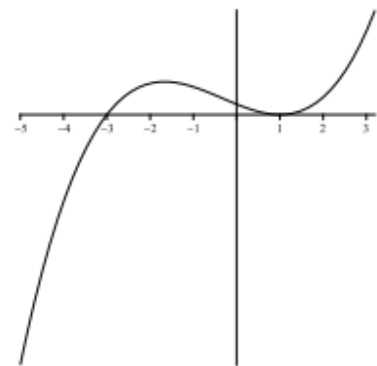
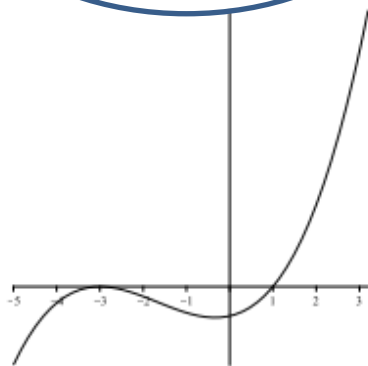
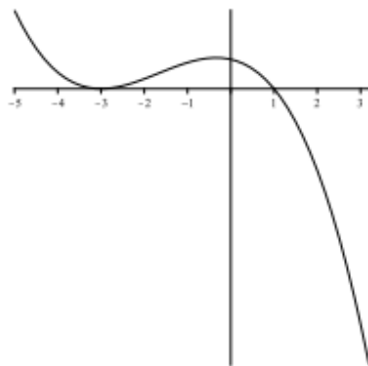
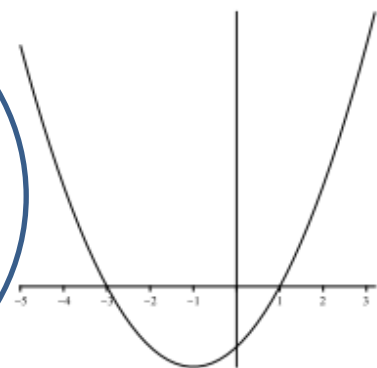
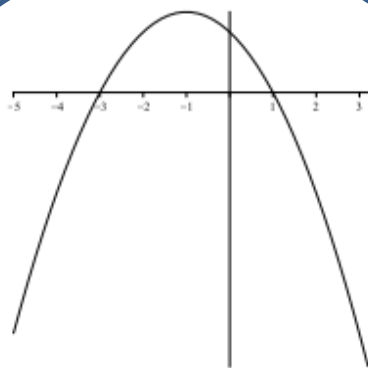
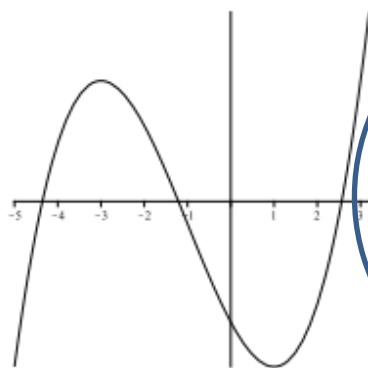
$f(x)$ has a domain of $(0, \infty)$.

$f'(x)$ has a domain of $(0, \infty)$.

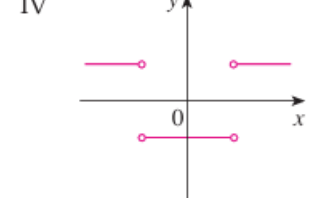
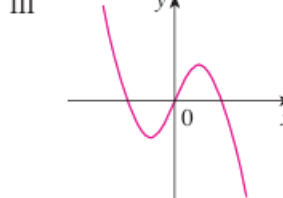
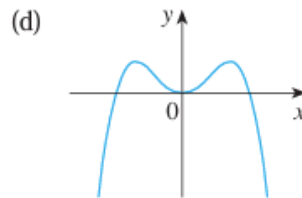
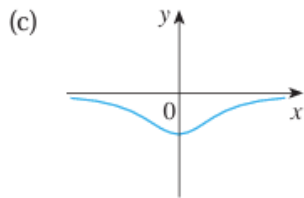
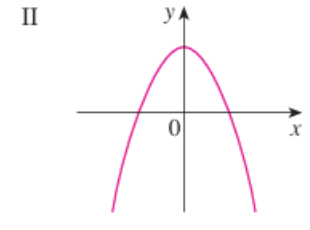
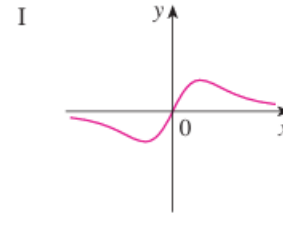
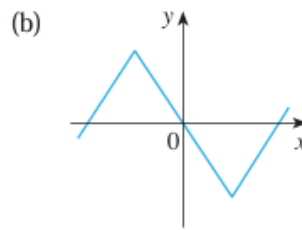
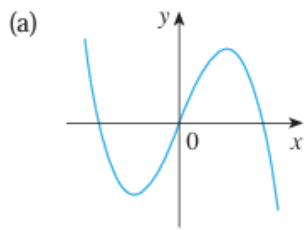
2. The graph of the function $f(x)$ is shown below.



One of the six graphs below is the graph of $f'(x)$. Circle the graph of $f'(x)$.

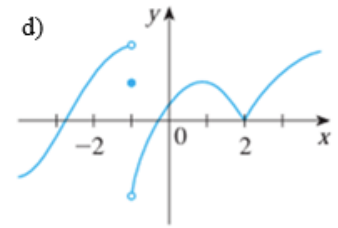
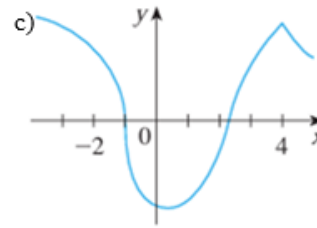
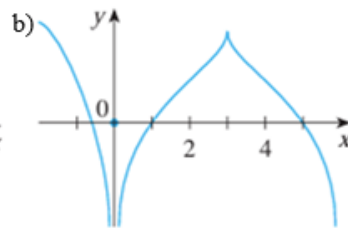
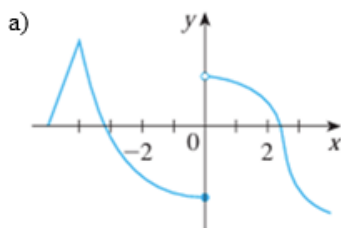


3. Match the graph of each function in (a) – (d) with the graph of its derivative in I-IV.



- a) We match this graph with II.
 b) We match this graph with IV.
 c) We match this graph with I.
 d) We match this graph with III.

4. The graph of f is given. State, with reasons, the numbers at which f is not differentiable.



- a) f is not differentiable at $x = -4$ & 0 .
 At $x = -4$ there is a corner point and not tangent line can be drawn.
 At $x = 0$ the function has a jump discontinuity and therefore no tangent line can be drawn to model the behavior of the curve on both sides of $x = 0$.
- b) f is not differentiable at $x = 0$ & 3 .
 At $x = 0$ there is an infinite discontinuity and therefore the function is not defined. So, no tangent line can be drawn at $x = 0$.
 At $x = 3$ the function has a cusp. No tangent line can be drawn to model the behavior of the curve on both sides of $x = 3$.
- c) f is not differentiable at $x = -1$ & 4 .
 At $x = -1$ the function appears to have a vertical tangent line here which would have an undefined slope.
 At $x = 4$ the function has a cusp. No tangent line can be drawn to model the behavior of the curve on both sides of $x = 4$.
- d) f is not differentiable at $x = -1$ & 2 .
 At $x = -1$ the function has a jump discontinuity and therefore no tangent line can be drawn to model the behavior of the curve on both sides of $x = -1$.
 At $x = 2$ the function has a cusp. No tangent line can be drawn to model the behavior of the curve on both sides of $x = 2$.

5. Suppose that we know that $g(x)$ is an *even* function that is differentiable at $x = 1$. we know that $g(x)$ has the following values:

$$g(-4)=1 \quad g(-2)=1 \quad g(-1)=3 \quad g(0)=3 \quad g(2)=1 \quad g(3)=1$$

Find $\lim_{x \rightarrow 1} g(x)$.

Since $g(x)$ is even then $g(x) = g(-x)$. Since $g(x)$ is differentiable at $x = 1$ then $g(x)$ is continuous at $x = 1$.

So, $\lim_{x \rightarrow 1} g(x) = g(1) = g(-1) = 3$.

6. TRUE or FALSE: If a function $f(x)$ is continuous at $x = c$, then $f'(x)$ must exist.

This is false. A counter example here would be $f(x) = |x|$ at $x = 0$. $f(x) = |x|$ is continuous at $x = 0$ but is not differentiable there and therefore, $f'(0)$ does not exist.